

Cell Type	Bacterial, gram positive	Molecules Electroporated	DNA: plasmid pLAR33, 18kB, supercoiled.
Species Used	<i>Lactobacillus</i> sp., strain 100-33		

Before the Pulse

Cell Growth Medium	MRS broth (DIFCO)	Growth Phase at Harvest	O.D. (600) =0.6 to1.0 (log)
		Pre-pulse Incubation	none

Wash Solution 2% glycerol in 10 mM NaPO₄ (pH 6.0)

The Pulse

Instruments Used Gene Pulser® apparatus

Electroporation Temperature 0 °C

Electroporation Medium 2% glycerol in 10 mM NaPO₄ (pH 6.0)

Cell Density 100x growth when harvested

Volume of Cells 50 µl

DNA Concentration 1 µg

DNA Resuspension Buffer Not given

Volume of DNA 1 to 5 µl

Cuvette Gap 0.2 cm

Voltage 2.5 kV

Field Strength 12.5 kV/cm

Capacitor 25 µF

Resistor (Pulse Controller) 200 Ω

Time Constant 1 to 3 msec

After the Pulse

Outgrowth Medium MRS broth

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

We have found that the electroporation of plasmids into our intestinal *Lactobacilli* isolates is highly strain-dependent. Our efficiencies range from 0 to 2 X 10⁽⁵⁾.

Outgrowth Temperature 37 °C

Length of Incubation 1 to 3 hrs, depends on selection.

Selection Method or Assay Used Antibiotic resistance (erythromycin)

Electroporation Efficiency 2 X 10⁽⁵⁾ transformants / µg DNA

Per Cent Survival 1%

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Survey Number

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